Code No: 181AP **R22**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, March/April - 2023
PROGRAMMING FOR PROBLEM SOLVING
(Common to CSE(AI&ML), CSE(IOT), AI&DS, AI&ML)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

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PART - A

(10 Marks)

1.a)	Define Algorithm.	[1]
b)	What is Compiler?	[1]
c)	Write about self referential structures.	[1]
d)	Demonstrate about strcpy() function.	[1]
e)	Demonstrate fclose() function.	[1]
f)	Write about #ifdef in brief.	[1]
g)	What is recursion?	[1]
h)	Explain about function signature.	[1]
i)	What is searching?	[1]
j)	Define Big-O notation.	[1]

PART - B

(50 Marks)

2.a)	Explain bitwise operators in detail.	
b)	What is a flowchart? Give the symbols/shapes used in the flowchart.	[5+5]
	OR	
3.a)	Explain in detail about Loop Control Statements in C.	
b)	Distinguish between formatted and unformatted statements.	[5+5]
4.a)	Give syntax for Two Dimensional Arrays and Explain in detail.	
b)	Explain String conversion functions with examples.	[5+5]
	OR	
5.a)	Write a program on Structure Bit Fields and explain in detail.	
b)	Explain types of Pointers in brief.	[5+5]
6.a)	Explain file operations in detail.	
b)	Write a C program to implement feof() and ferror() functions.	[5+5]
	OR	
7.a)	Discuss about Pre-processor commands.	
b)	Explain text file modes in detail.	[5+5]

8.a) Explain categories of user defined functions.
b) Write about limitations of the recursive functions. [5+5]

OR

9.a) Explain pointer for inter function communication.
b) Write a program to find factorial of a given number using recursion. [5+5]

10.a) Write a program for insertion sort.
b) What is linear search? Explain the linear search procedure with following numbers to search 77.
"88, 22, 38, 29, 44, 12, 77, 81, 98" [5+5]

OR

11.a) Distinguish between linear search and binary search with an example.
b) Write a program for bubble sort. [5+5]

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